Badrikanath Praharaj

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B.Tech '26 · IISc robotics researcher · digital twins of places humans shouldn't go · 7 ICRA/RSS/NeurIPS workshop papers

Work Experience

† *Dr. Pavankumar Tallapragada (Assoc. Prof., IISc)*

- **Kosha Labs (Founding Engineer)** *Sept 2026 - Present*
  Increased QC-passed video by **23.5%** in datasets for **OpenAI** and **Google DeepMind** by decoupling gap-filling from segment fusion in a dynamic-programming segmenter and tuning thresholds on blind human labels. Built a blur detector fusing gyro angular velocity and Laplacian variance via logistic regression: recall **34%→68%**, **59%** fewer false rejects.
- **Indian Institute of Science, Bangalore (Robot Learning Intern)** *Jun 2026 - Present*
  HiRo Lab, advised by **Prof. Ravi Prakash**. Extending a real-to-sim pipeline that converts real-world datasets into **Isaac Sim** digital twins for RL benchmarking; built **2** environments. Created a **DAgger** framework in which a motion-capture-based expert labels states the learner visits, training an exploration policy for uncertainty-aware SLAM.
- **Indian Institute of Science, Bengaluru (Control and Optimization Intern)**† *Aug 2025 - June 2026*
  Designed a real-time multi-robot coordinator to bridge planning and safe execution, using quadrant-token mutual exclusion, arrival-time arbitration, and wait-for-graph cycle detection; conflict-free, scalable to 10 robots, validated across 125+ optitrack experiments, maintaining spatio-temporal accuracy with 0.25s schedule error and 0.12s inter-robot wait.
- **Intel, Bangalore (LLM Inference & Optimization Intern)** *May 2024 - Sept 2024*
  Cut T5-small CPU inference latency from **120ms to 84ms (-30%)** with OpenVINO quantization and operator fusion. Fine-tuned it on CNN/DailyMail summarization using LoRA and layer-wise LR decay, raising **ROUGE-L by 4.2%**.

Published Research

- **MineXplore: A RL Exploration Benchmark for GPS-Denied Environment (ICRA 2026@Xplore)** [arxiv]
  Addressing the absent mining-RL benchmark, built an open-source MuJoCo simulation of a **real Chilean copper mine** via a contour-to-MJCF pipeline, achieving **95.38%** geometric IoU and validating PPO policies at **88.9% coverage**.
- **Subterra: A Validated Benchmark for GNSS-Denied Tunnel Navigation (RSS 2026 @WPCIS)** [link]
  Lacking a real-geometry tunnel-RL testbed, rebuilt a **351 m** metro tunnel from LiDAR scans as a MuJoCo POMDP, validated at **97.1%** fidelity (**2.46 cm** surface error) across 199 metrics, where a PPO agent reached **90.02%** coverage.
- **Teaching Robots to Say 'I Don't Know': SENTINEL for Uncertainty SLAM (ICRA 2026@UOWR)** [arxiv]
  Engineered a **training-free** reliability layer flagging silent LiDAR failures on sub-$300 robots, fusing scan geometry with LiDAR/RGB-D depth consistency to reject corrupted scans via reliability scores, validated across **5 failure modes.**
- **CADENCE: Predicting MAPF Execution Time Beyond Sum of Costs (ICRA 2026@MARS)** [arxiv]
  Built a **480-trial robotic hardware corpus** (7 robots) to predict completion time, showing primitive motion burden cuts held-out error by **48.6–59.8% MAE** over Sum-of-Costs via scenario-held-out ridge and mixed-effects models.
- **Reasoning & Evaluation of Visual Evidence with Aligned Language (NeurIPS 2025@WiML)** [arxiv]
  Fine-tuned forgery detectors fail to generalize; reframed detection as zero-shot VLM prompting via **eight-factor scoring and 3×3 region grids**, lifting Gemini's Columbia F1 **118%** (**0.39→0.85**) and GPT-4.1 to **0.92**, rivaling baselines.

Projects

†*IISc Bangalore, rest VIT Bhopal*

- **MineSwarm: Multi-Robot Mining Digital Twin** [github] *Jun 2025 – Present*
  Engineered a **multi-agent RL digital twin** (MADDPG+TD3, PyBullet) for robust decentralized navigation of 3 LiDAR-equipped robots in a GPS-denied mine, using curriculum learning, phase-aware replay, and twin-critic stabilization. Reached **91% goal-success** across an 8-phase difficulty curriculum, scaling zero-shot to real-world hardware.
- **PowerSplit: Endurance-Optimized Multi-Robot Coordination** † [github] *Aug 2025 – Sept 2025*
  Built 3 coordinated robots where a high-level planner computes plans, lightweight onboard controllers run velocity control, and a split battery isolates and right-sizes compute and motor power, extending runtime from 65 to 74 minutes (+13.5%).

Skills & Tools

- **Expertise**: Digital Twins & Real-to-Sim Reconstruction, Sim-to-Real Transfer & Fidelity Validation, Synthetic Data Generation, Multi-Agent RL, Multi-Robot Coordination, Reliability-Aware Perception, LLM Fine-Tuning & Post-Training
- **Tools**: MuJoCo, Isaac Sim, PyBullet, ROS 2, OptiTrack, LiDAR & RGB-D, PyTorch, LoRA, OpenVINO, Python, Git

Education

Vellore Institute of Technology, Bhopal *Sept 2022 - Sept 2026*
B.Tech in Electronics and Communication (AI and Cybernetics) *CGPA: 8.9/10*

Achievements

- **Conference Reviewer**: Evaluating submissions in the Robotics track of **NeurIPS 2026 @WIML** & MEXAAIDC.
- **Department Topper**: Scored a perfect **10.0 GPA** in final year, and ranked **1st** overall in 1st, 2nd & 4th years.
- **LycheeAI x REVEL Hackathon**: **55+ public votes** won our Isaac Sim pick-and-place build the Community Favorite.
- **Eureka Club Co-Founder**: Co-lead a **10+ member** R&D team in a 70+ club; guided **7+ papers** & 1 patent.